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Manifold embeddings achieve comparable performance with multispectral imagery for time-series based land disturbance detection

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ABSTRACT

Dense long-term time series multispectral imagery is crucial for monitoring Earth's surface and detecting disturbances in near-real-time. However, the massive storage requirements of such data pose significant challenges. Dimensionality reduction techniques have been widely applied in remote sensing to address the curse of dimensionality, yet achieving lossless recovery of multispectral data across global locations and varying surface conditions remains difficult. Additionally, it is unclear whether reduced features retain temporal continuity and can integrate effectively with time series algorithms for disturbance detection. This study leverages Uniform Manifold Approximation and Projection (UMAP) for multispectral dimensionality reduction, trained on Harmonized Landsat Sentinel-2 (HLS) imagery. The resulting manifold embeddings are applied to the Continuous Change Detection and Classification (CCDC) algorithm for land disturbance detection. Two key findings emerge: (1) We developed a general UMAP-based dimensionality reduction model that works across global seasons, with manifold embeddings preserving time series coherence and exhibiting stable value ranges. (2) The embeddings achieved comparable performance to full-spectrum multispectral data in image prediction and disturbance detection with CCDC. This research highlights the potential of manifold learning to efficiently store and process dense satellite imagery while maintaining the ability to detect diverse land disturbances.

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KEYWORDS
Land disturbances detection; time-series; dimension reduction; anomaly detection; manifold learning

1. Introduction

1.1. Background

In recent years, terrestrial remote sensing observation has undergoing a paradigm shift from change detection to remote sensing monitoring (Woodcock et al. 2020). The

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availability of dense time series remote sensing data has significantly advanced the development of earth observation technologies (Suh, Zhu, and Zhao 2024). A series of methods based on dense time series data has become a crucial means for monitoring surface disturbances. These methods have been widely applied in monitoring wildfires (Fotakidis et al. 2023), floods (Damle and Yalcin 2007), pest outbreaks (Ye et al. 2021), deforestation (Bullock, Woodcock, and Olofsson 2020), agricultural expansion (Zabel et al. 2019), and construction changes (Suh, Zhu, and Zhao 2024). Compared to change detection algorithms based on two temporal phases, methods utilizing dense time series data offer the advantage of continuous Earth monitoring, enabling more timely and effective detection.

However, dense time series data requires substantial data storage and incurs high computational costs. For instance, processing a Landsat ARD image using the COLD algorithm requires 1000–3000 h, and storing the time series imagery requires 0.5–2 terabytes (Zhu et al. 2020). Such demands present a substantial barrier to achieving continuous surveillance and prompt detection of extensive surface disturbances. Looking ahead, an increasing number of satellites with higher spectral resolutions and an expanded range of spectral bands are slated for launch. For instance, the Landsat Next mission, planned for launch in late 2030/early 2031, will boast an impressive 26 spectral bands (Wulder et al. 2022). The proliferation of data intensifies the challenges associated with the all-bands approach to land disturbance detection. Therefore, it is imperative to simplify vast amounts of time series multispectral data while minimizing data loss to accommodate various types of complex surfaces.

Currently, most studies focus on the continuous monitoring of specific types of surfaces, which can be achieved by selecting sensitive bands or constructing specific indices (Bullock, Woodcock, and Olofsson 2020; Deng and Zhu 2020; Ye et al. 2021). However, the Earth's surface is highly complex, with significant variations in the sensitive wavelength ranges of different land cover types. Thus, employing full-spectrum multispectral data for disturbance detection is crucial. Developing a multispectral dimensionality reduction model suitable for a variety of complex global surfaces, capable of achieving lossless compression and recovery of any spectrum, is key to addressing this issue. Can current dimensionality reduction techniques attain this objective? Do the reduced features maintain continuity within the time series? Are they applicable for disturbance detection in complex substrates? What is the efficacy of such detection? These questions remain unanswered.

1.2. Related work

Dimensionality reduction methods. These methods yield a reduced set of features that encapsulate the full spectral information, thereby offering a more efficient alternative to analyzing each individual spectral band. Numerous advanced dimensionality reduction techniques are now available, including linear methods like Principal Component Analysis (PCA) (Maćkiewicz and Ratajczak 1993), Linear Discriminant Analysis (LDA) (Xanthopoulos, Pardalos, and Trafalis 2013), Independent Component Analysis (ICA) (Comon 1994), and nonlinear methods like Locally Linear Embedding (LLE) (Roweis and Saul 2000), t-distributed Stochastic Neighbor Embedding (t-SNE) (Maaten 2008), Auto-Encoder (Rumelhart, Hinton, and Williams 1988), UMAP

(McInnes, Healy, and Melville 2018), though their comparative performance for being applied to time series analysis is still unknown. These techniques can condense extensive spectral information into a more manageable space while effectively preserving the original data, especially the nonlinear methods. Utilizing these compressed features in time series model construction obviates the need to select specific bands or indices for different land cover types and significantly reduces data storage requirements and computational costs. UMAP, a high-dimensional clustering technique based on manifold learning developed in recent years, stands out among dimensionality reduction algorithms due to its lossless compression capabilities and rapid computation speed (Allaoui, Kherfi, and Cheriet 2020). It has been widely applied across various fields, achieving exceptional results (Ali et al. 2019; Becht et al. 2019; Dorrity et al. 2020; Yu et al. 2023). In the realm of remote sensing, UMAP has been utilized for characterizing spectral blending spaces, adept at capturing nuanced spectral differentiations among land cover categories (Small and Sousa 2022; Sousa and Small 2022). As a nonlinear dimensionality reduction technique, UMAP offers significant advantages over PCA and t-SNE for compressing multispectral data in time-series analysis. PCA, constrained by its linear projection, can only capture global variance directions and thus often misses complex, nonlinear spectral relationships. t-SNE, while highly effective at preserving local neighborhood structure, incurs substantial computational cost and breaks global topology, hindering its use on large datasets. In contrast, UMAP efficiently preserves both local neighborhoods and the overall manifold topology in a single low-dimensional embedding, while achieving significantly faster runtimes than t-SNE (Becht et al. 2019). Moreover, unlike PCA and t-SNE – which require scene-specific transformations (PCA via per-scene eigenvectors; t-SNE via affinity graphs with no built-in out-of-sample method) – UMAP learns a single mapping that generalizes directly to any new multispectral image without recalculating parameters or storing per-image coefficients. This unified, reusable model, together with UMAP's high computational efficiency, makes it ideally suited for large-scale, long-term remote sensing workflows that demand stable, scalable, and repeatable spectral compression and reconstruction. This unified, reusable model, together with UMAP's high computational efficiency, makes it ideally suited for large-scale, long-term remote sensing workflows that demand stable, scalable, and repeatable spectral compression and reconstruction. Moreover, it ensures the comparability of features across different images, which is crucial for preserving temporal patterns. Although Autoencoders offer powerful nonlinear dimensionality reduction capabilities, their optimization is primarily driven by reconstruction accuracy, which often leads to representations that emphasize overall data fidelity rather than discriminative features relevant to spectral variation. Moreover, the latent features extracted by Autoencoders frequently lack physical interpretability. In contrast, UMAP is designed to preserve the nonlinear structural relationships among spectral bands, enabling the compression of high-dimensional multispectral data into a few compact and representative components. This not only reduces dimensionality but also retains key information more relevant to change dynamics. Therefore, this study adopts UMAP for the compression and reconstruction of multispectral data, which is subsequently integrated with time-series analysis for effective disturbance detection.

Time series methods for land disturbance detection. The central objective of land disturbance monitoring via satellite time series analysis is to develop models that capture the

natural variability and trend of the Earth's surface. In the past decades, many time series-based methods to detect land disturbances has been developed, with representatives including Breaks For Additive Seasonal and Trend (BFAST) (Verbesselt et al. 2010), LandTrendr (Kennedy, Yang, and Cohen 2010), long short-term memory (LSTM) (Graves and Alex 2012), CCDC (Zhu and Woodcock 2014b), Continuous Monitoring of Land Disturbance (COLD) (Zhu et al. 2020), Stochastic Continuous Change Detection (S-CCD) (Ye, Rogan, Zhu, and Eastman 2021), Object-based Continuous Monitoring of Land Disturbance (OB-COLD) (Ye, Zhu, and Cao 2023) and other deep learning-based methods (Mao et al. 2024; Haixu He et al. 2024). BFAST and LandTrendr are capable of identifying disturbances in historical data, but they lack predictive capabilities and cannot perform timely online disturbance detection. LSTM captures long-term dependencies and complex patterns for prediction and real-time detection but faces high computational complexity, tuning challenges, and reduced performance with very long sequences. CCDC utilizes harmonic models derived from dense time series satellite data to depict natural surface fluctuations, thereby adeptly monitoring long-term land cover changes. Its forecasting abilities facilitate real-time updates and seamless online surveillance. The other algorithms are improvements derived from CCDC, offering enhancements in accuracy, timeliness, and object-oriented analysis.

1.3. Contribution

The central question this paper endeavors to explore is whether reduced features can optimally preserve original information and temporal patterns. Additionally, it examines whether these features have the potential to supplant multispectral data for time series analysis and surface disturbance detection, thereby streamlining the monitoring process. To address this issue, the paper undertakes the following tasks:

(1) A general multispectral dimensionality reduction model was trained based on UMAP, demonstrating strong spectral recovery and generalization capabilities. Multispectral projections from any location, time, or type globally exhibit stable shapes and structures within this low-dimensional manifold space. This ensures that embeddings derived from any image projection are comparable, maintaining temporal continuity in reduced features obtained from images at different times. The time series embeddings retain temporal patterns, avoiding uncertainty in the value ranges of reduced features across different images;
(2) The manifold embeddings were applied to a time series algorithm (specifically the CCDC algorithm in this study) for predicting images across arbitrary seasons and detecting four different types of surface disturbances. The results indicate that the low-dimensional manifold embeddings hold significant potential to replace full-spectrum multispectral data in dense time series analysis.

1.4. Outline

This paper is structured as follows: Section 2 outlines the methods employed in this study; Section 3 presents the data used in the research, and the two-dimensional manifold

space representing global surface spectra, evaluating the predictive capacity of UMAP embeddings in conjunction with the time series algorithm CCDC, and the disturbance detection capabilities of the manifold embeddings both qualitatively and quantitatively; Section 4 discusses the robustness and interpretability of the embedding space, demonstrates two UMAP embeddings perform on par with higher-dimensional embeddings, analyzes the contributions of each UMAP embedding within the time series, and articulates the advantages and limitations of the proposed method; finally, Section 5 summarizes the findings of this study.

2. Methodology

The flowchart of this study is shown in Figure 1. Our method was divided into the following three steps. Firstly, each time-series multispectral image is projected into the two-dimensional manifold space using the UMAP model trained on globally representative land surface spectra dataset, i.e. UMAP_TRAIN, constructing two new time series of the transformed features (Section 2.1). Next, we apply the CCDC algorithm to build stable curves for the time series of the two features, facilitating feature prediction at any given time point (Section 2.2). The UMAP model reconstructs the multi-spectral image at the current time based on the predicted two features. The reconstructed multi-spectral image is considered to represent the current state of the land surface. Finally, comparisons

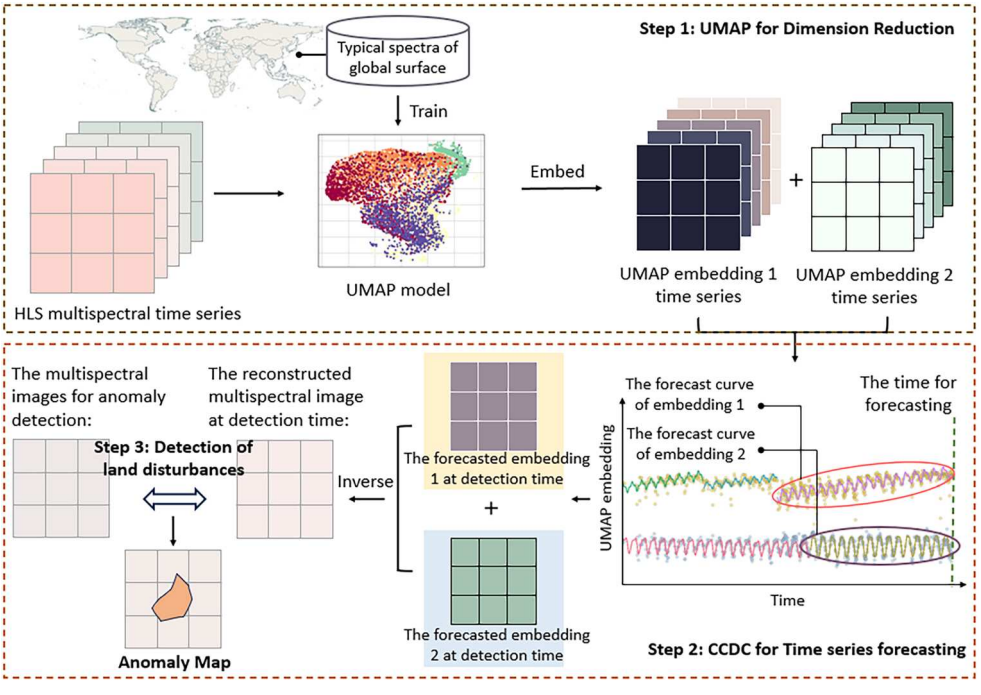


Figure 1. The workflow of the proposed method. Our method was divided into three steps. For step 1, the UMAP model is trained using globally typical spectra, and the manifold spatial features of two time series are projected from multispectral images. For step 2, the features at detection time are predicted by CCDC, and the multispectral image is reconstructed through UMAP inverse projection. For step 3, disturbances are detected by comparing simulated and real images.

are made between the input image to be detected and its reconstructed counterpart across multiple spectral bands to monitor disturbances (Section 2.3).

2.1. UMAP for dimension reduction

UMAP is a nonlinear dimensionality reduction algorithm that incorporates advanced mathematical theories including Riemannian geometry and topological algebra. It is founded on three assumptions about the data: (1). the data is uniformly distributed on Riemannian manifold; (2). the Riemannian metric is or can be approximated as locally constant; (3). the manifold is locally connected. The implementation of UMAP mainly involves the following three steps:

(1) A function is formulated to depict the relationship between two points in a high-dimensional space, e.g. 6 bands form a 6-dimensional space, as shown in the following equation:

$$p_{j|i} = e^{-\frac{d(x_i,\ x_j) - \rho_i}{\sigma_i}} \tag{1}$$

$$k = 2^{\sum_i p_{ij}} \tag{2}$$

$$p_{ij} = p_{i|j} + p_{j|i} - p_{i|j} \cdot p_{j|i} \tag{3}$$

where p_{ij} represents the joint probability of points i and j being neighbors in the high-dimensional space. x_i and x_j represent two points in the high-dimensional space that may be neighbors. $p_{j|i}$ represents the probability of sample point j occurring given sample point i. $d(x_i,\ x_j)$ represents the distance between sample points i and j. ρ_i is the distance from point i to its nearest neighbors. σ_i is determined by k, where k indicates how many points in high-dimensional space point i can be neighbors with. By adjusting k, the optimal σ_i can be determined. $d(x_i,\ x_j)$ can be set to any distance function.

(2) A function is formulated to depict the relationship between two points in a low-dimensional space, as shown in the following equation:

$$q_{ij} = (1 + a(y_i - y_j)^{2b})^{-1} \approx \begin{cases} 1, & y_i - y_j^2 \le \min_dist \\ e^{-y_i - y_j^2 - \min_dist}, & otherwise \end{cases} \tag{4}$$

where q_{ij} represents the joint probability of points i and j being neighbors in low-dimensional space. y_i and y_j represent two points in low-dimensional space that may be neighbors. Parameters a and b can be adjusted to determine the clustering of data in low-dimensional space. However, in practical applications, *min_dist* is the hyperparameter that users adjust to determine the clustering of data.

(3) Stochastic gradient descent is employed for parameter optimization to make p_{ij} and q_{ij} as close as possible. The loss function is utilized, defined as follows:

$$C(X, Y) = \sum_i \sum_j \left[p_{ij}(X) \log\left(\frac{p_{ij}(X)}{q_{ij}(Y)}\right) + (1 - p_{ij}(X)) \log\left(\frac{1 - p_{ij}(X)}{1 - q_{ij}(Y)}\right) \right] \quad (5)$$

where $C(X, Y)$ represents the value of the cross-entropy loss function. X and Y represent the distances between two points in high-dimensional and low-dimensional space, respectively. $p_{ij}(X)$ and $q_{ij}(Y)$ represent the probabilities of two points becoming neighbors in high-dimensional and low-dimensional space, respectively.

In this study, we utilized UMAP to achieve a streamlined spectral representation, effectively minimizing data storage requirements and computational costs. A UMAP model was trained using the UMAP_TRAIN dataset, which contains ten thousand randomly selected typical spectra derived from HLS images of 227 scenes. Subsequently, we projected all the time series of multispectral images through the trained model into two UMAP features. These two features' time series, replacing the multispectral time series, are used for CCDC forecasting.

2.2. CCDC for time series prediction

The CCDC algorithm (Zhu and Woodcock 2014b) is used for time series forecasting. Firstly, the time series of surface reflectance (SR) images are processed into continuous and clear observations by filtering out clouds and shadows using the Function of mask (FMASK) (Zhu, Wang, and Woodcock 2015) and MultiTemporal mask (TMASK) (Zhu and Woodcock 2014a) algorithms. It is worth noting that FMASK employed the QA band, while TMASK's input of the green and SWIR2 bands was substituted with two UMAP features in this paper. Next, the full CCDC model (Zhu et al. 2015) is used to fit the time series observations and predict the reflectance or features at any given moment. The function is as follows:

$$\begin{aligned} \hat{p}(i, x) = {} & a_{0,i} + a_{1,i} \cos\left(\frac{2\pi}{T} x\right) + b_{1,i} \sin\left(\frac{2\pi}{T} x\right) + a_{2,i} \cos\left(\frac{4\pi}{T} x\right) \\ & + b_{2,i} \sin\left(\frac{4\pi}{T} x\right) + a_{3,i} \cos\left(\frac{6\pi}{T} x\right) + b_{3,i} \sin\left(\frac{6\pi}{T} x\right) + c_{1,i} x \\ & \{\tau^*_{k-1} < x \leq \tau^*_k\} \end{aligned} \quad (6)$$

where x is Julian date. i is the ith HLS band ($(i = 1,\ 2,\ 3,\ 4,\ 5,\ \ and\ 6)$) or features. T is number of days per year ($T = 365.25$). $a_{0,i}$ coefficient for overall value for the ith HLS band or UMAP features. $a_{1,i}$ and $b_{1,i}$ are coefficients for intra-annual change. $a_{2,i}$ and $b_{2,i}$ are coefficients for intra-annual bimodal change. $a_{3,i}$ and $b_{3,i}$ are coefficients for intra-annual trimodal change. $c_{1,i}$ is coefficient for inter-annual change (slope). τ^*_k is the kth break points. $\hat{p}(i, x)$ represents predicted value for the ith Landsat Band or UAMP features at x Julian date. The model can effectively capture both the multimode intra-annual cycles (harmonic coefficients) and the inter-annual trend (a constant and slope term). Least absolute shrinkage and selection operator (LASSO) is used to fit the function. The full model requires at least 24 observations to be initialized, and the total time span must exceed one year.

For the thresholds to detect disturbances, the CCDC algorithm considers that the sum of the squares of the normalized changes deviates follows a chi-square distribution with

degrees of freedom equal to the number of bands used for detection. The threshold for determining disturbances is defined by the following formula (Zhu et al. 2020):

$$\sum_{i=1}^{n}\left(\frac{\rho_i-\widehat{\rho_i}}{RMSE_i}\right)^2 \sim \chi^2(n) > \chi^2_{0.99} \tag{7}$$

where i represents the input band, while n represents the total number of bands. In this study, n is 6. ρ_i and $\widehat{\rho_i}$ is the observed value and predicted value for ith HLS band. $RMSE_i$ represents the root mean square error between the full model and the observed value for the ith band. We adopted the default change magnitude threshold, i.e. $\chi^2_{0.99}$, for our CCDC comparative experiments.

In this study, we employed CCDC for image forecasting to achieve our objective: utilizing historical spectral data as a knowledge base to synthesize images at any given time that conform to the natural dynamics of the Earth's surface. Following the initial preprocessing of the time-series images for each study area, we reduced the dimensionality of the multi-band time series images by projecting them into two UMAP feature time series images, thereby optimizing storage efficiency.

2.3. Land disturbance detection

To identify land disturbance events, we utilized the fitted CCDC model to forecast two UMAP feature images at the desired time point. Given the constraint of a limited number of UMAP features for statistical analysis, i.e. two features, formula (7) was not directly employed to identify disturbances. Instead, we projected the two predicted UMAP features back into the original 6-band space to reconstruct a synthetic image for the specific time of interest. This synthetic image represents the expected land dynamics at that time, assuming no disturbances had occurred. Subsequently, we identified disturbances by comparing the observed 6-band image with the synthetic one. A disturbance event is detected when the difference between the observed and synthetic images exceeds a threshold defined by the following equation:

$$P_{\text{disturbance}} = \begin{cases} 0, & \frac{1}{n}\sum_{i}^{n}|b_i-\widehat{b_i}| < t \\ 1, & \frac{1}{n}\sum_{i}^{n}|b_i-\widehat{b_i}| \geq t \end{cases} \tag{8}$$

where $P_{\text{disturbance}}$ signifies whether the pixel represents land disturbance. A value of 0 means no land disturbance occurred at the pixel, while a value of 1 indicates land disturbance. i represents the input band, while n represents the total number of bands. In this study, n is 6. b_i represents the pixel values from the ith band of the actual imagery, while $\widehat{b_i}$ represents the pixel values from the ith band of the synthetic imagery. t represents reflectance threshold for determining whether land disturbance has occurred in this study. The default value for t is 0.04. The choice of such threshold has been justified and detailed in supplementary material. It is important to note that clouds will not be identified as disturbances; instead, they are filtered out using the Quality Assessment (QA) band of the HLS data. Furthermore, in this study, we confirmed a disturbance event based on a single observation, necessitating the setting of the 'conse'

parameter to 1 within the CCDC code. The source code, including this parameter setting, can be accessed at https://code.usgs.gov/lcmap/pyccd.

2.4. Accuracy assessment

In this study, we first evaluated the UMAP model by comparing the reconstructed reflectance spectra with the observed ones, which assess the accuracy of UMAP projection and back-projection processes, ensuring that the UAMP features keep the all-band information. The metrics used for this evaluation are the coefficient of determination R^2 and the root mean square error (RMSE).

$$R^2 = 1 - \frac{\sum (y_i - \widehat{y_i})^2}{\sum (y_i - \bar{y})^2} \tag{9}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum (y_i - \widehat{y_i})^2} \tag{10}$$

where y_i are the observed values, $\widehat{y_i}$ are the predicted values, and $\bar{y}$ is the mean of the observed values. n is the number of observations.

In this study, quantitatively assess the precision of the proposed method in disturbance detection, we visually interpreted 1400 sample points across five study areas, covering a total area of approximately 20000 square kilometers. The principle for validating pixel samples is as follows: in areas with extensive disturbances, we generate a higher number of random validation samples to ensure the collection of an adequate quantity of disturbance examples. Due to the broader reach of flood and wildfire-affected regions, 400 samples were allocated to each, while the smaller areas impacted by landslides and construction received 100 samples each. Furthermore, within each region, we applied stratified random sampling across both disturbed and undisturbed areas. We vectorized the disturbed regions. The area of undisturbed regions is more extensive than that of disturbed regions, so among the 1400 samples, 1070 correspond to undisturbed surfaces, while 330 pertain to disturbed areas. This paper calculates five metrics: precision, recall, false alarm rate, F1-score, and accuracy. The formulas are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{11}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{12}$$

$$\text{falsealarm} = \frac{FP}{FP + TN} \tag{13}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

$$A = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

where TP represents true positives, FP represents false positives, FN represents false negatives, and TN represents true negatives.

3. Experimental results and analysis

3.1. Data and pre-processing

3.1.1. Dataset for embedding training

To establish a globally robust spectral manifold space, we compiled an image dataset that encompassed a diverse range of land surface types, covering all inhabited continents during all meteorological seasons. Since our study relies on dense time series analysis, we therefore utilized the Harmonized Landsat and Sentinel-2 (HLS) image dataset, which provides higher temporal resolution and satisfactory spatial resolution compared with the with the use of either Sentinel-2 or Landsat imagery in isolation. We selected locations and seed values identical to the study of Michael Schmitt et al. (2019), where a curated SEN12MS dataset was built. Finally, we compiled a total of 227 HLS images: 55 for spring scenes, 58 for summer scenes, 75 for fall scenes, and 39 for winter scenes. Each image spans an area of roughly 12000 square kilometers. The locations of the images for the four seasons are depicted in Figure 2. Figure 3 displays representative image scenes and the corresponding sampling points selected within each scene.

Two sets of spectral data were extracted from these images, each containing ten thousand spectra. Through experimentation, we determined that this quantity was optimal for training our model (detailed in 3.2). The first set of spectral dataset, called UMAP_TRAIN, was randomly sampled from cloud-free regions of high-quality images, ensuring that the proportions of spectra for various land cover types match their actual distribution on the Earth's surface. This strategy is considered to better represent the land surface, and this dataset was used to train the UMAP model. The second dataset, called UMAP_VAL, was extracted through manual interpretation of different land cover types. In total, we interpreted 2052 spectra for developed areas, 2123

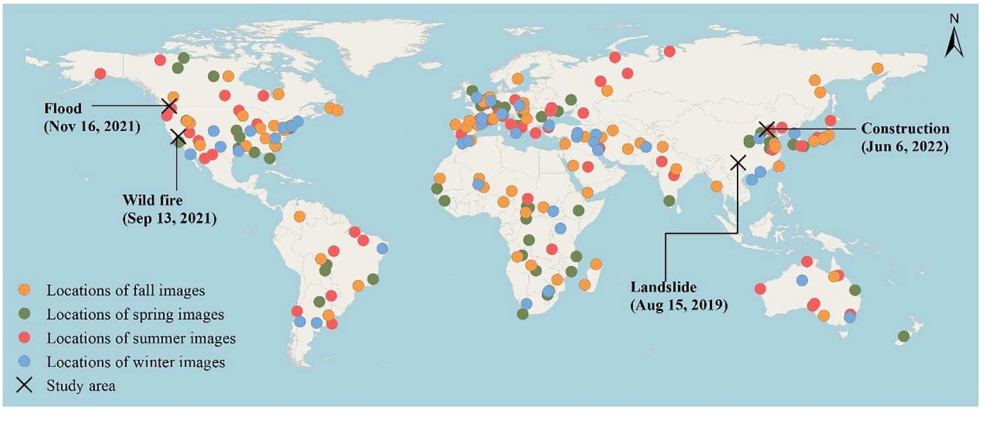


Figure 2. Locations of 227 HLS images for extracting land surface spectra in each season, 55 for spring scenes, 58 for summer scenes, 75 for fall scenes, and 39 for winter scenes. Four selected sites for anomaly monitoring of land disturbance events in this study.

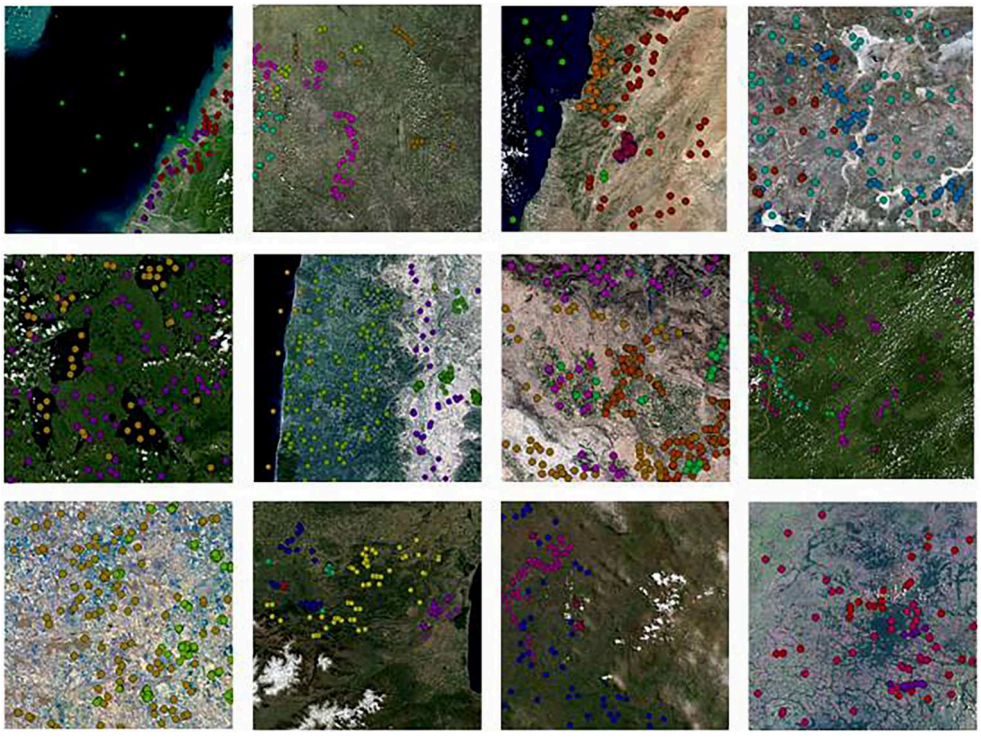


Figure 3. Examples of HLS images with points representing the locations where spectra were extracted. Five types of spectra were extracted in total: developed land, bare land, farmland, vegetation, and water.

spectra for farmland, 1953 spectra for bare land, 3079 spectra for vegetation, and 1532 spectra for water. We followed these rules to extract spectra for each land surface: Developed areas primarily included impermeable surfaces like roads and building rooftops; Bare land included exposed natural surfaces like rocks, sandy areas, and soil without vegetation; Farmland included both cultivated and uncultivated agricultural land; Vegetation includes all land surfaces with vegetation signals, excluding farmland; Water bodies included both inland and near-shore waters. In this paper, the first dataset was used to train the UMAP model, while the second dataset was used to evaluate performance of the UMAP model.

3.1.2. Dataset for disturbance detection

To validate the proposed disturbance detection method, we chose five influential disturbance events across the world, including three natural disasters and one human-induced land surface modification. These events include flooding (which occurred in early November 2021 in parts of British Columbia, Canada, and Washington State, USA), wildfires (which occurred in August 2021 in California, USA, and in January 2020 on Kangaroo Island in South Australia), landslides (which occurred in July 2019 in Guizhou Province, China), and construction (which took place in March 2022 in Hebei Province, China). We used HLS images with dense time series for disturbance detection and processed all historical image tiles from the past three years for each of

the five aforementioned study areas. The HLS dataset comprises two subsets: L30 and S30, each with distinct band ranges. We selected the bands common to both sensors for disturbance detection.

3.2. Embedding space of global multispectral data

Figure 4 shows the embedding effect of UMAP_VAL and UMAP_TRAIN onto the model trained with UMAP_TRAIN. The parameter settings for model training are provided in the supplementary material. The gray points in the figure correspond to projections from the UMAP_TRAIN, while the colored points, distinguished by various land cover types, reflect projections from the UMAP_VAL. The overlap between projections from two entirely distinct multispectral datasets attests to the ability of our trained UMAP model to represent a wide range of global spectral types within this two-dimensional space. In this two-dimensional manifold space, spectra from various global land cover types exhibit stable shapes and structures. The distinct clustering of different land cover types demonstrates that the low-dimensional manifold space preserves the characteristics of land surface in high-dimensional space. The low-dimensional space still retains the ability to distinguish various land cover types. However, various land cover types still exhibit mixing. Cropland and vegetation are significantly mixed due to their similar spectral characteristics. Cropland and bare soil also show substantial mixing, primarily because fallow cropland shares spectral features with soil. In our visual interpretation, fallow cropland is categorized within cropland. Soil, bare rock, and unused land are collectively classified as bare soil. Moreover, bare soil is significantly mixed with developed areas, as certain streets and rooftops exhibit spectral characteristics similar to bare soil. In essence, the manifold space we have constructed exhibits stable topological structures, allowing spectra from various global types to be projected onto this space and undergo lossless reconstruction. In other words, this model can effectively condense HLS multispectral data into two dimensions, and this process is entirely reliable. Moreover, these two features reveal the differences between land cover types and their transitions, providing mechanistic insights. They are optimal indicators for detecting surface disturbances.

Figure 5 showcases the performance of spectral projections from various land cover types onto a two-dimensional manifold space using models trained with different quantities of spectra. Models trained with different quantities of spectra vary in size. When the number of spectra used to train the model is fewer than 10000, the morphology of spectra projected onto a two-dimensional space differs from that of models trained with more than 10000 spectra. When trained with more than 10000 spectra, the model consistently projects spectra into a stable two-dimensional space, maintaining a form that is simply mirrored horizontally. The model's size increases with the growing number of spectra used in training. Taking into account both the stability and size of the model, we ultimately chose the model trained with 10000 spectra.

Figure 6 shows the reconstruction performance of the UMAP embedding trained in this study on various types of spectra. The figure illustrates that irrespective of the spectrum type, our trained model can nearly flawlessly reconstruct the original multiband spectra using only two UMAP embeddings.

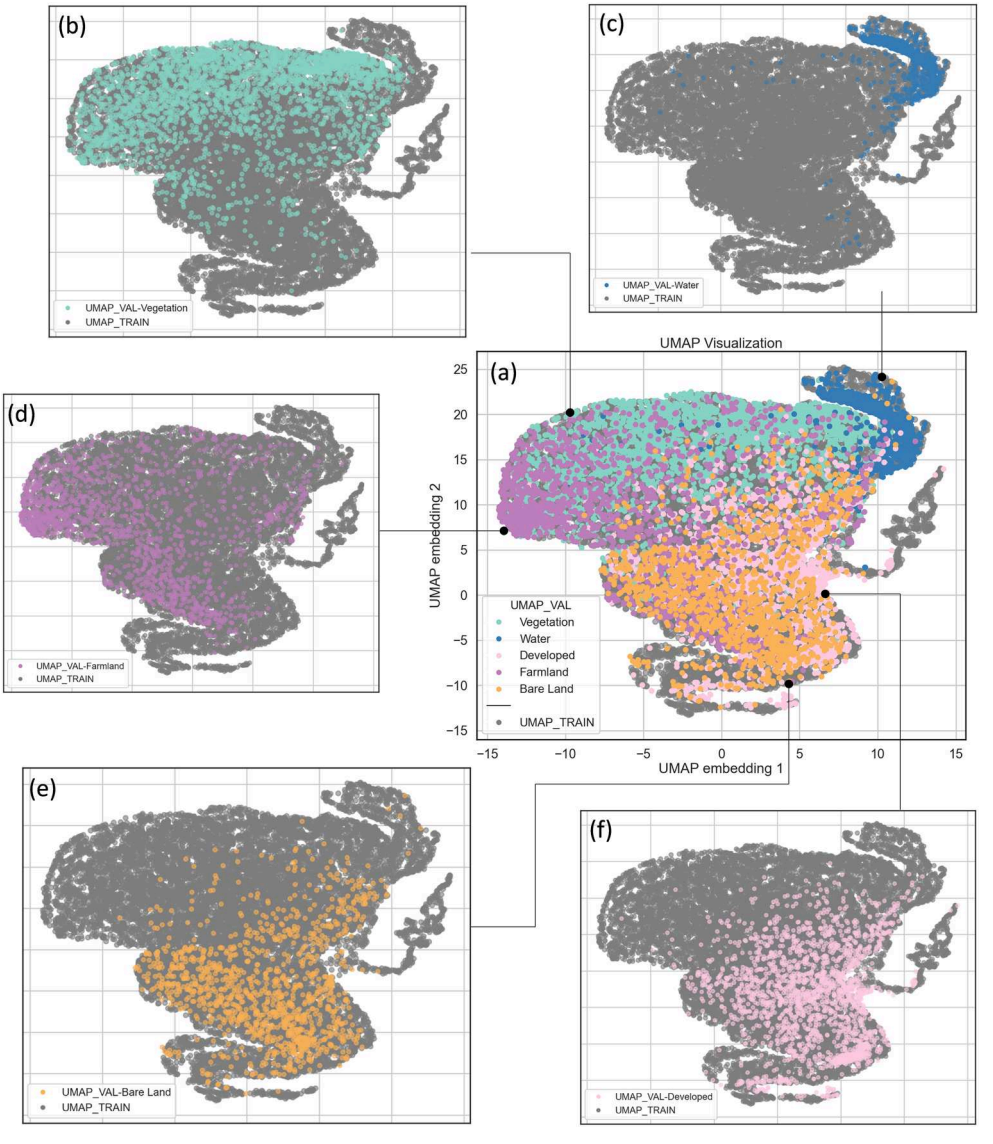


Figure 4. (a) The morphology of the UMAP_TRAIN and the UMAP_VAL for different land cover types in two-dimensional manifold space. (b) – (f) respectively represent the morphology of vegetation, water, farmland, bare land, and developed areas in two-dimensional manifold space.

3.3. Performance of UMAP embeddings in CCDC

Figure 7 illustrates the synthetic images predicted using two UMAP embeddings, compared with the corresponding real images. The first case study showcases a region in Guizhou, China, characterized by a subtropical monsoon climate, featuring distinct seasons. There are discernible seasonal fluctuations in vegetation growth. The second case study showcases a region in western Washington state, United States, characterized by a temperate maritime climate. The area experiences stable temperatures and precipitation year-round, with mostly cloudy and rainy weather and consistent vegetation cover.

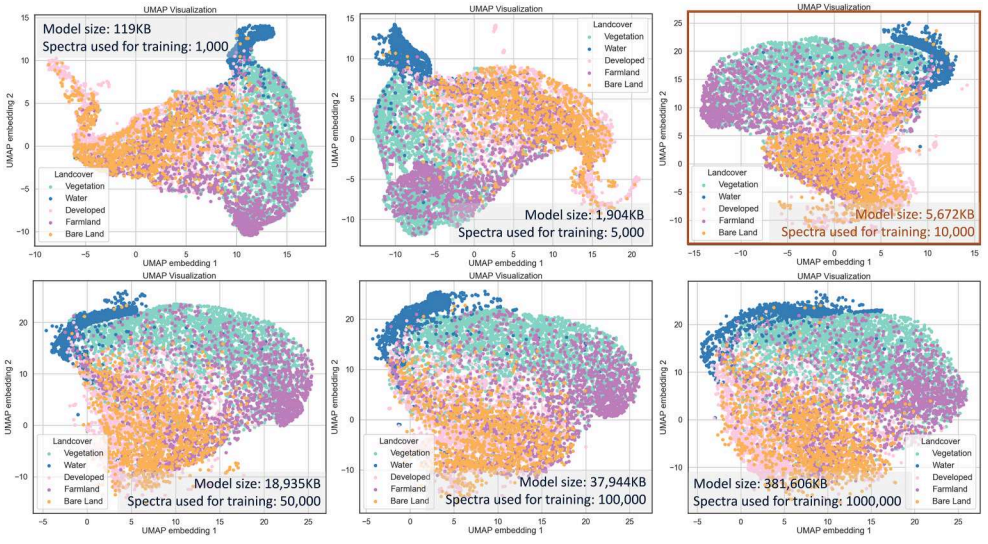


Figure 5. The performance of spectral projections from various land cover types onto the two-dimensional manifold space using models trained with different quantities of spectra. The models are trained using 1000, 5000, 10000, 50000, 100000, and 1000000 spectra, respectively. The other parameters are configured with 'min_dist' set to 0.8 and 'n_neighbors' set to 8. The two-dimensional manifold space represented on the brown axes is projected by the model used in this study, trained with 10000 spectra.

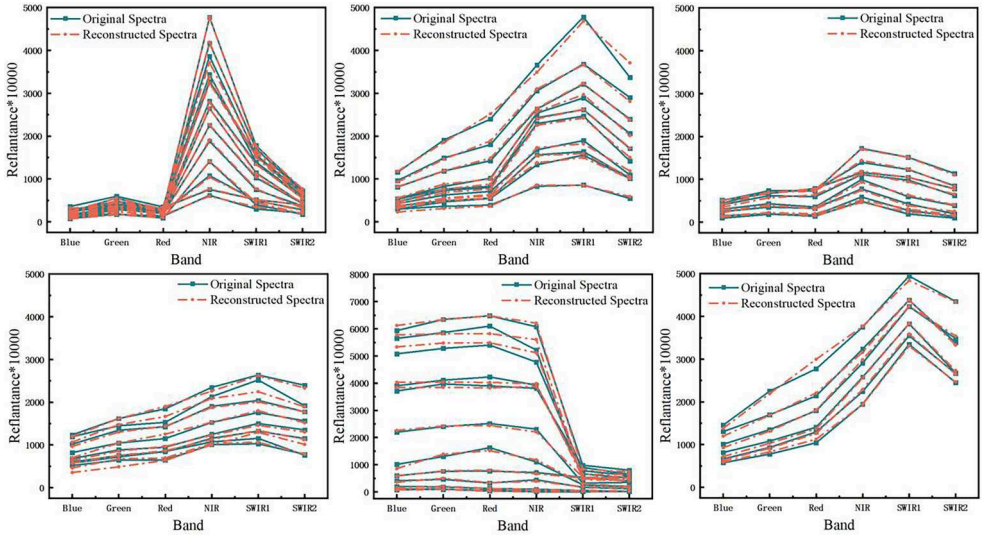


Figure 6. The reconstruction performance of the UMAP embedding trained in this study on various types of spectra.

Our synthetic images achieve a striking resemblance to authentic imagery and CCDC adeptly simulates areas where the seasonal variations in vegetation are less pronounced. In essence, CCDC skillfully models the temporal patterns of UMAP features in time

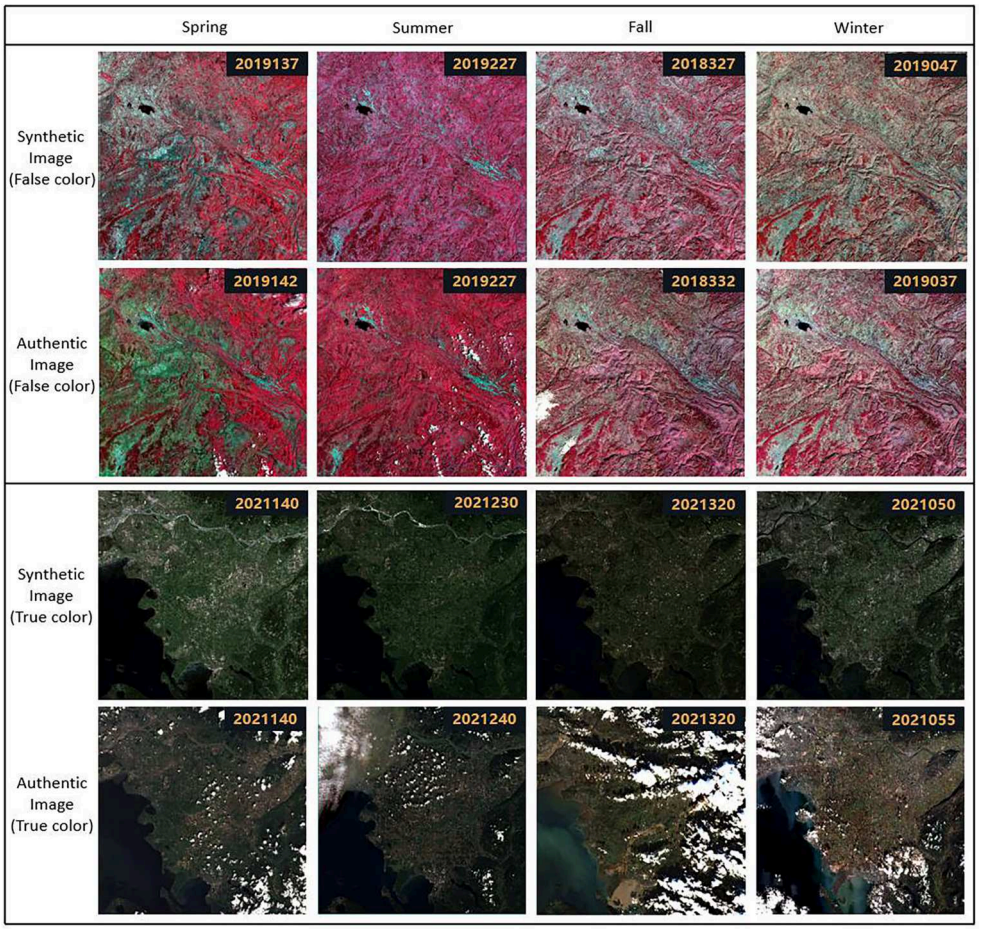


Figure 7. The comparison between synthetic images generated by the method proposed in this article and authentic images captured across four seasons. Example 1: Location in Guizhou, China – False-color representation. Example 2: Location in Washington State, USA – True-color representation. The labels in the top-right corner of the images represent Julian dates, indicating the year and the day of this year. Due to cloud interference, the authentic images displayed may represent images captured a few days prior to or following the dates of the synthetic images.

series and achieves relatively accurate predictions. The predicted features can be faithfully restored to the original bands through the UMAP model, precisely forecasting imagery regardless of the season or land cover type. Combining CCDC with UMAP offers a significant advantage in time series forecasting: it operates on just two features derived from the data, rather than across all spectral bands. This approach substantially reduces both storage and computational demands, making it particularly beneficial for applications involving extensive long-term time series imagery. The storage requirement has been reduced from six bands to two. and the computation time has also been significantly reduced. For a time series of 370 points, the average time to predict two UMAP feature bands is 0.0195 s, while predicting all bands requires 0.0620 s. The time needed to predict all six bands of an image is approximately 3.2 times that required to predict just two bands.

Figure 8 shows the fitting performance between the values predicted directly by CCDC for each band and the values for each band reconstructed from UMAP embeddings predicted using CCDC. A total of 500000 pixels from the four study areas were used for fitting. According to our validation results, UMAP exhibited commendable reconstruction accuracy for the two short-wave infrared bands, the green band, and the red band, with slightly diminished performance for the near-infrared and blue bands. Overall, the band values directly predicted by CCDC and those reconstructed from the UMAP feature values predicted by CCDC exhibit a high degree of correlation and minimal error. The R^2 values are all above 0.7, and the RMSE remains below 0.04.

3.4. Accuracy of disturbance detection

Using the method outlined in our paper, we detected disturbances in the five study areas by analyzing time-series HLS images captured over the three years leading up to the event. To ensure a balanced distribution of disturbed and undisturbed areas for reliable random point sampling during validation, the size of each study area was determined based on the spatial extent of the disturbance, ranging from approximately 60–12000 square kilometers. For example, landslides typically affect small regions, requiring smaller validation areas, whereas wildfires and floods often cover large areas, necessitating broader spatial coverage. Each selected region comprised approximately 300 satellite images to support time series analysis. Figure 9 presents the pre – and post-disturbance HLS images, the reconstructed post-disturbance images and the disturbance map detected by the proposed method for the five cases. The synthetic images perform well

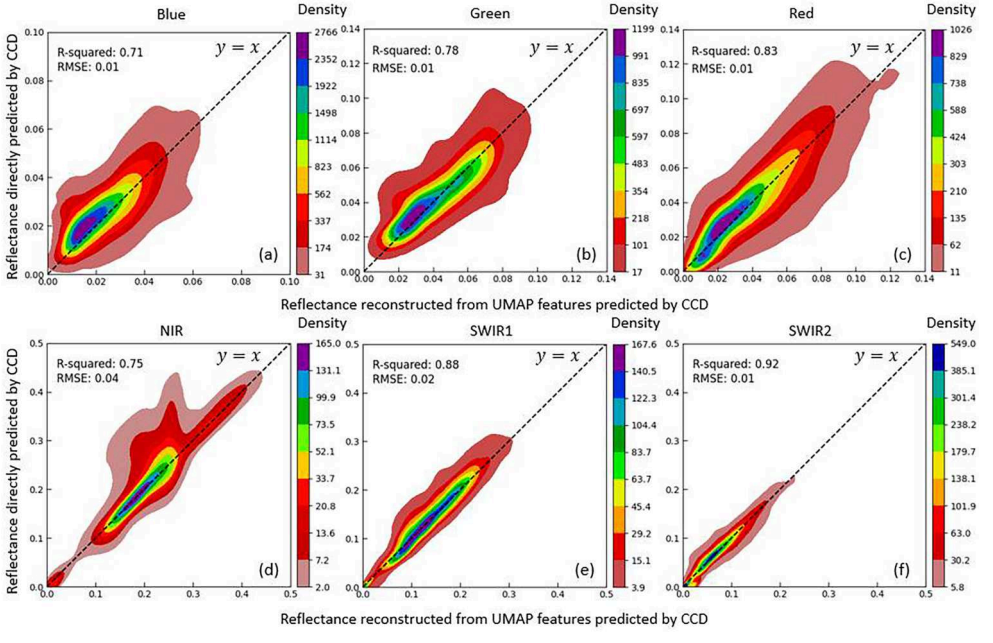


Figure 8. Comparison of reflectance values obtained from direct CCDC predictions and CCDC-predicted UMAP feature reconstructions. (a–f) Performance of two methods in reflectance synthesis of blue, green, red, NIR, SWIR1 and SWIR2.

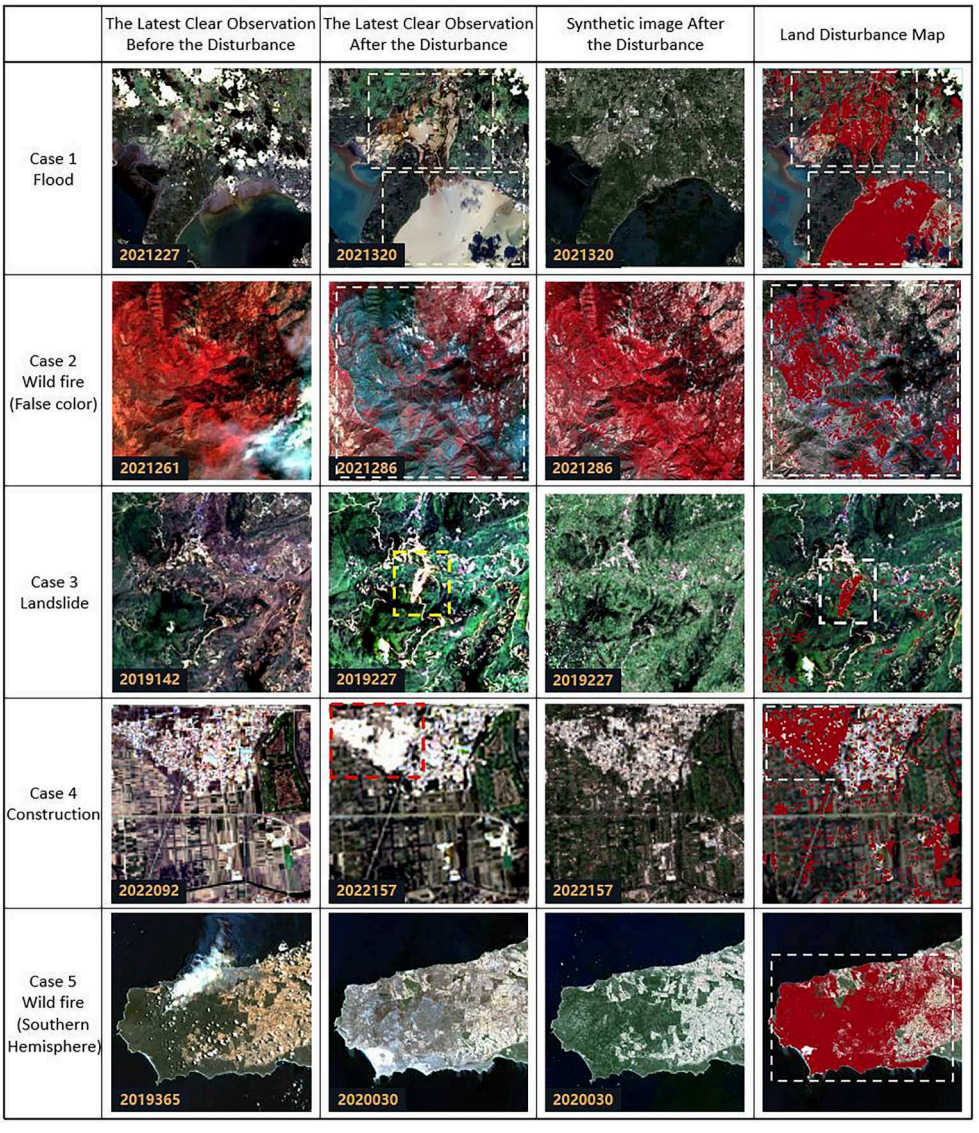


Figure 9. Land disturbance mapping for five cases by the proposed method in this paper. The first column presents clear image obtained before the disturbance, taken closest in time to the disturbance's occurrence. The second column shows clear image obtained after the disturbance, taken closest in time to the disturbance's occurrence. The third column presents synthetic images of the same day as the second column, reconstructed using historical image features through the methodology proposed in this paper. The fourth column shows the mapping of land disturbance. The red areas are the identified disturbances. The dashed boxes in the second and fourth columns indicate the areas affected by the disturbance. The labels in the bottom left corner of the images in the first three columns represent the Julian date, indicating the year and the day within that year.

whether displayed in true color or false color. In certain cloudy and rainy areas, like Case 3, clear images are difficult to obtain. Due to the considerable time gap between the clearest image obtained prior to the disturbance and the image at the time of detection, there's a substantial difference in vegetation coverage. Traditional change detection methods are

Table 1. The algorithm accuracy comparison for the two methods.

	Disturbance Detection with Manifold Embeddings	Disturbance Detection with All Multispectral Bands
Precision	0.66	0.54
Recall	0.56	0.81
False alarm	0.09	0.22
F1-score	0.61	0.65
Accuracy	0.83	0.77

inadequate for precise disturbance identification. Synthetic images, forecasted based on historical trends, successfully predict the surface conditions during disturbance occurrence. While some instances of over-detection persist in the disturbance identification results, they primarily involve small, fragmented patches.

Table 1. illustrates the accuracy of both methods. The precision and false alarm rate of CCDC with manifold embeddings are superior to those with all bands; however, the recall rate is lower when using manifold embeddings. This indicates that using manifold embeddings compromises a certain degree of recall in order to mitigate the false alarm rate. Overall, the F1-score and accuracy are similar, indicating that both methods perform comparably in disturbance detection. This indicates that using manifold embeddings maintains detection accuracy even with fewer spectral bands, significantly reducing redundant data storage and unnecessary computation time. Neither all bands nor manifold embeddings achieved high precision in disturbance detection, as we used only a single observation, indicating near-real-time monitoring. This accuracy represents the normal level for near-real-time detection using the CCDC algorithm. To improve precision, increasing the frequency of observations is recommended (Ye, Zhu, and Suh 2024). Figure 10 illustrates the performance of these two methods across two examples.

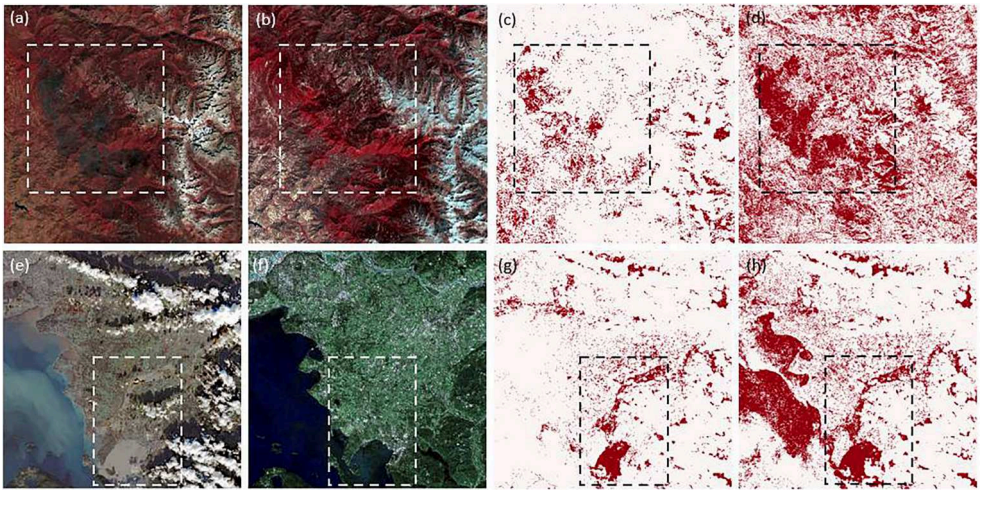


Figure 10. (a) and (e) depict authentic HLS images of two distinct disturbance scenes. (a) depicts a false-color representation of a fire scene, while (b) shows a true-color representation of a flood scene. (b) and (f) depict synthetic images of two distinct disturbance scenes by the proposed method. (c) and (g) are disturbance maps generated using the proposed method in this article. (d) and (h) are disturbance maps generated by the CCDC algorithm using only one time step to identify disturbances in this article.

From Figure 10, it is evident that for certain regions, employing the CCDC algorithm with only a single observation for disturbance detection may inevitably lead to issues of over-identification. Thus, multiple consecutive observations are considered necessary. The manifold embeddings method shows slightly fewer instances of over-identification, suggesting the potential to reduce the required number of observations for effective disturbance detection in the future, enabling a more efficient implementation.

4. Discussion

4.1. Robustness and interpretability of embedding space

In this study, we endeavor to develop a robust spectral compression model that can project spectra from any global region, at any time, and of any type into a low-dimensional space while enabling their precise reconstruction. This approach aims to minimize data storage requirements. In the realm of remote sensing, traditional dimensionality reduction techniques include PCA, Tasseled Cap Transformation, and Minimum Noise Fraction (MNF). PCA projects high-dimensional data into a lower-dimensional space by transforming the coordinate system, resulting in low-dimensional features with minimal correlation. This ensures that a few components can encapsulate the majority of the information contained in the high-dimensional data. PCA can be interpreted as a linear model that performs the transformation between high-dimensional and low-dimensional data. However, the limitations of PCA are as follows: (1) It necessitates dimensionality reduction for the entire image, rendering the independent transformation of individual pixels unfeasible. PCA is unable to produce a universally applicable model independent of the current input data, leading to limited flexibility. (2) Each feature is uninterpretable and contains a different amount of information. Thus, pixels from different times and regions are not inherently comparable, making PCA unsuitable for time series analysis (Dong and Qin 2018). The tasseled cap transformation reduces dimensionality by converting multispectral data into a few physically meaningful components using fixed coefficients. While it serves as a universal linear transformation model for the same sensor, its conversion relationships are excessively simplistic and uniform. Furthermore, the number of transformed components is still substantial, offering no substantial advantage over using the original multispectral data for time series disturbance detection.

In addition to traditional dimensionality reduction methods, deep learning models such as Autoencoders (Hinton and Salakhutdinov 2006) can be trained to achieve dimensionality reduction for multispectral data, incorporating both linear and nonlinear transformation factors within the model. Its limitation is that it compresses spectra aggressively, yielding low-dimensional features that lack physical meaning. UMAP, on the other hand, is a dimensionality reduction technique based on manifold learning. It can not only train a general model to compress spectra of any type, time, or region with effective reconstruction of the original spectra but also produce UMAP features that have physical significance. Figure 11 illustrates that when projecting any two images with complex surface cover types into a two-dimensional manifold space, the scatter points not only fall within the same range but also retain a similar overall distribution. Dividing these scatter points into several regions reveals that the points in each region correspond to pixels of the same land cover type in the images. Thus, we conclude

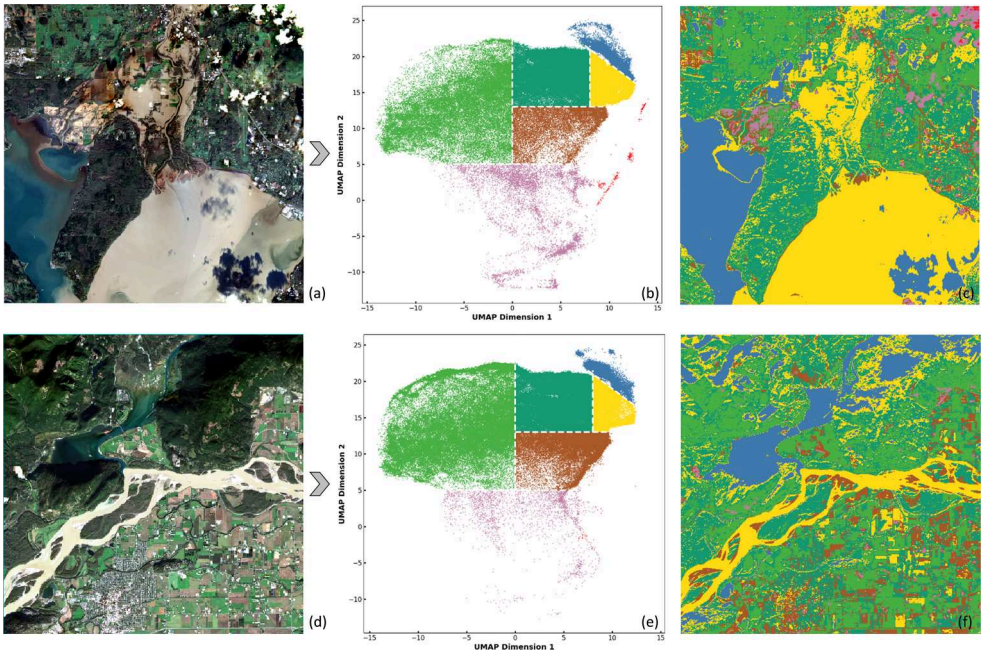


Figure 11. (a) and (d) are two arbitrary images with complex land cover types. All pixels from these images are projected into a two-dimensional manifold space, as shown in (b) and (e). (b) and (e) are divided into several regions, and the scatter points in these regions correspond to pixel locations in the images, as illustrated in (c) and (f).

that while individual UMAP features lack the tangible physical meaning of tasseled cap transformation features, pairs of UMAP features can collectively represent different types of land cover. Land cover types can be represented by the coordinate positions of two UMAP features.

4.2. Evaluating the optimal UMAP dimension

Figure 12 presents the results of disturbance detection using different numbers of UMAP embeddings. To minimize the storage required for time series analysis, the multispectral data were compressed into two UMAP embeddings. Despite this aggressive reduction, the two embeddings are sufficient for effective disturbance detection, showing performance comparable to that of using three, four, or five embeddings. Whether in the reconstructed multispectral images from prediction or in the disturbance maps, the two UMAP embeddings even achieve results comparable to those obtained with five embeddings.

4.3. Contribution of UMAP embeddings in CCDC

While UMAP features faithfully reconstruct reflectance across various spectral bands, their ability to maintain temporal patterns is equally vital for land surface characterization. Therefore, we investigated how time-series UMAP features perform within the CCDC algorithm. Figure 13 illustrates the performance of the CCDC model on the time series of each band in HLS data and each UMAP feature. In fact, the fitting

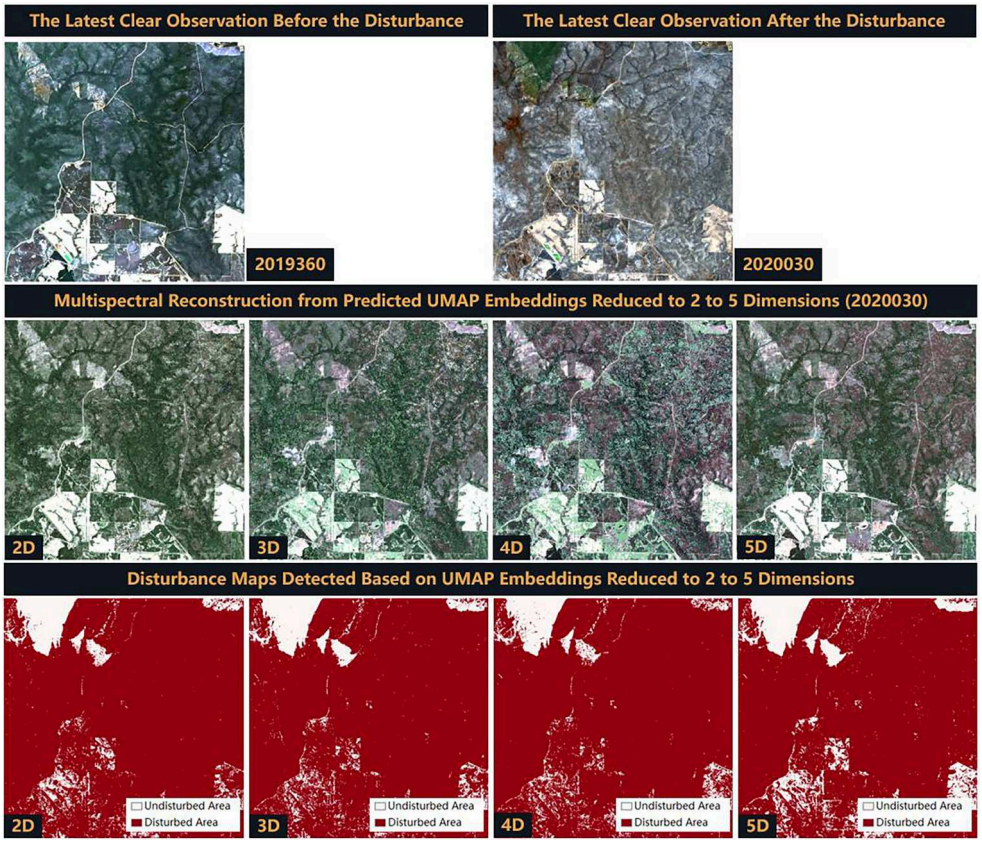


Figure 12. The two images in the first row represent the most recent observation before the fire and the first available observation after the fire. The seven-digit numbers on the right indicate the date of acquisition, with the first four digits representing the year and the last three digits indicating the day of the year (DOY). The four images in the second row are multispectral reconstructions obtained by performing time series prediction using UMAP embeddings reduced to 2, 3, 4, and 5 dimensions, respectively. These represent predicted post-fire conditions. The four images in the third row show the disturbance detection results based on UMAP embeddings reduced to 2, 3, 4, and 5 dimensions.

performance of CCDC varies across different bands, with discrepancies even in the breakpoints of the fitting. Overall, the detected breakpoint positions and total count are fixed, with certain breakpoints detectable in some bands while absent in others. The two compressed UMAP features also exhibit performance similar to these bands. From numerous experiments, we observed that the temporal continuity of UMAP embedding 1 (UMAP1) is slightly inferior, resulting in a higher error rate in detecting breakpoint positions, whereas UMAP embedding 2 (UMAP2) performs significantly better. This is attributed to another phenomenon we discovered: UMAP1 primarily reflects the overall trend changes in the time series, whereas UMAP2 provides a more detailed characterization of intra-annual variations. Overall, Both UMAP1 and UMAP2 demonstrate relatively good temporal continuity. Overall, despite UMAP1 and UMAP2 being derived from the nonlinear compression of various bands, they retain temporal continuity and are well-suited to the CCDC.

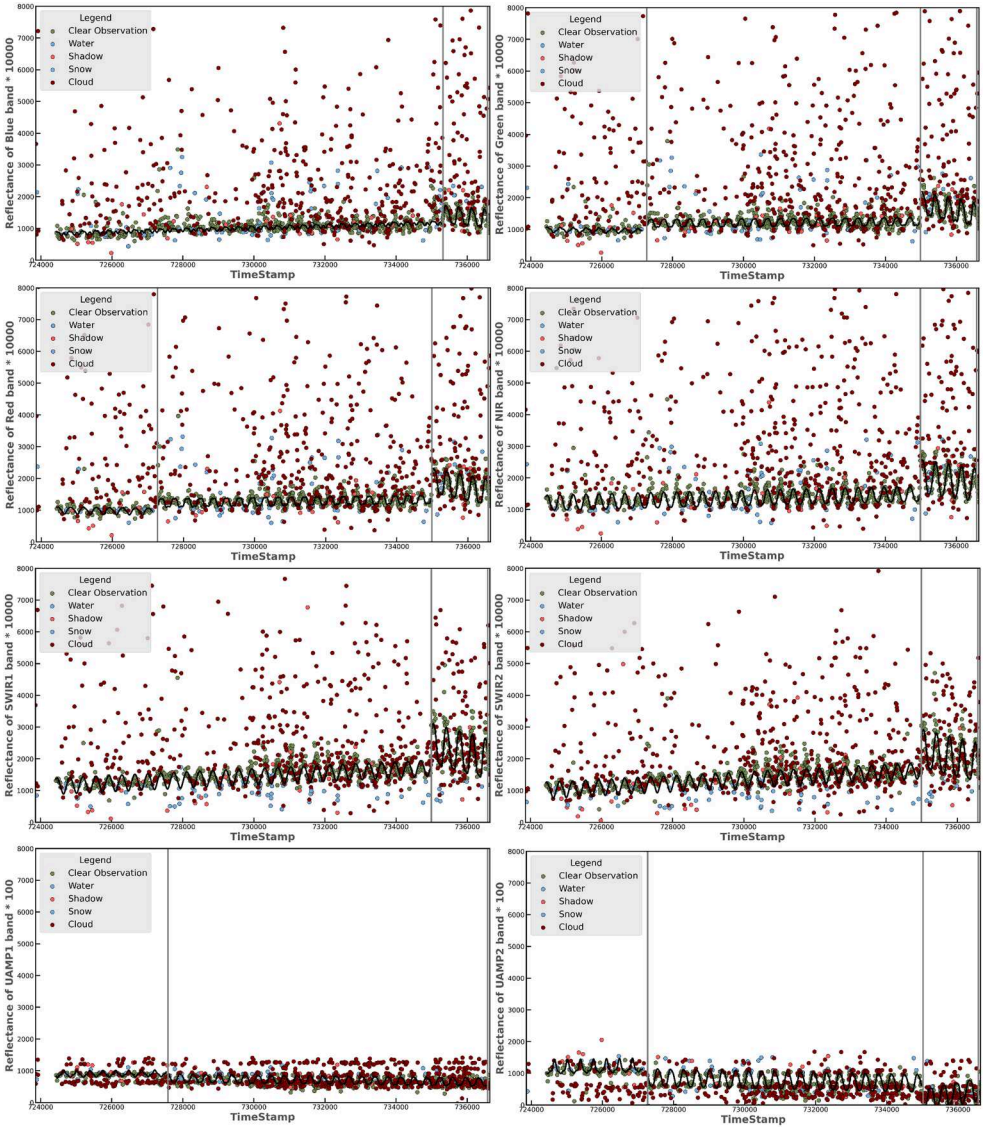


Figure 13. The curve fitting of the CCDC model to the time series observations of the six HLS data bands – Blue, Green, Red, NIR, SWIR1, and SWIR2 – as well as the two UMAP features. The scatter points represent the time series observations, which are categorized into five types based on the QA band of the HLS data: clear observations, water, shadow, snow, and clouds. Shadow and cloud data are excluded from the curve fitting process. The black curve represents the curve fitted by CCDC. The gray vertical lines indicate where the curve breaks.

4.4. Advantages and extensions

Using a UMAP model trained on globally representative spectra, HLS multispectral data is compressed into two-dimensional manifold features. Experiments have shown that the manifold features of any projected remote sensing image exhibit stable shapes and structures within the two-dimensional manifold space. This provides the theoretical

foundation for the comparability of features following multispectral image dimensionality reduction. Only when the features of different projected remote sensing images are comparable can they be reliably used for time series analysis. Manifold features have been proven to perform excellently in predictions and disturbance detection based on time series algorithm. Moreover, for disturbance detection targeting a specific land type, certain bands or indices can still yield impressive results. However, this approach lacks generalizability and cannot address all disturbance types. Different land features vary greatly in their sensitive wavelength ranges within the electromagnetic spectrum. For a single type of land surface, disturbance detection can be achieved by selecting the corresponding sensitive bands or constructing specific indices, which reduces dimensionality and storage space. However, this approach cannot accommodate the complex conditions of real-world surfaces. Therefore, dimensionality reduction methods that retain as much of the original spectral information as possible offer a greater advantage.

Indeed, the more abundant the spectral bands and the denser the time series in remote sensing imagery, the more conspicuous the advantages of our UMAP model become, signifying its broad and promising applications. Especially in the realm of hyperspectral remote sensing, the high dimensionality of data hampers the extensive utilization of hyperspectral imagery (Bioucas-Dias et al. 2013). Previous studies have compared the efficacy of various techniques for reducing the dimensionality of hyperspectral data and their application in hyperspectral classification. The findings reveal that, when compared with PCA, LDA, t-SNE, ICA, LLE, and other methods of dimensionality reduction, UMAP demonstrates superior precision and swifter processing. In the realm of hyperspectral image classification, it adeptly preserves the overarching features of images (Li et al. 2022). In the domain of time-series hyperspectral imagery, our algorithm holds profound significance, serving essential roles in target surveillance, disturbance detection, species invasion monitoring, plant phenotyping, and the assessment of wood moisture content and microbial decay in food through 'hyperspectral videos'.

4.5. Limitations

In this study, we chose HLS imagery for the experiments due to its denser time series, which lends itself well to continuous Earth observation and disturbance detection. However, this choice limits the UMAP model's full potential for lossless compression. Only six bands in HLS imagery support disturbance detection, resulting in a relatively low compression ratio. With more spectrally diverse remote sensing data, our UMAP model would better showcase its capabilities, achieving a more pronounced compression effect. This paper uses HLS imagery as an example to demonstrate that compression using the UMAP model can produce stable manifold spaces, with comparable features suitable for time series analysis.

5. Conclusion

To assess whether reduced features could replace multispectral data for time series analysis and surface disturbance detection, this study combines the advanced nonlinear dimensionality reduction algorithm UMAP with the time series algorithm CCDC to detect surface disturbance. First, we developed a manifold model for dimensionality

reduction, suitable for multispectral data of any land surface type, demonstrating robust stability and interpretability. Next, we applied the manifold embeddings within CCDC for disturbance detection, achieving accuracy comparable to that of full-band data. Based on the results and analyses, we made a few conclusions as follows: (1) Using manifold embeddings for disturbance detection streamlines the process while achieving optimal accuracy; (2) We no longer need to meticulously select sensitive bands or indices for specific applications or land cover types. The manifold features retain excellent multispectral properties and temporal patterns, making them fully adaptable to various disturbance detection scenarios; (3) Our approach offers promising potential, particularly with more comprehensive spectral data and denser time series.

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Geolocation information statement

The geolocation data in this article is authentic and sourced from actual measurements.

Author contributions

Conceptualization, M.L. and J.Q.; Data curation, M.L.; Formal analysis, M.L.; Funding acquisition, Q.W.; Investigation, Q.W.; Methodology, M.L., S.Y. and J.Q.; Project administration, Q.W.; Resources, Q.W.; Software, M.L.; Supervision, Q.W.; Validation, M.L.; Visualization, M.L.; Writing – original draft, M.L.; Writing – review and editing, M.L., J.Q. and S.Y. All authors have read and agreed to the published version of the manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data of this study are available from the corresponding author upon reasonable request.

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